UCLA Samueli
Chemical & Biomolecular Engineering

February 5, 2026

To: Distinguished Teaching Award Selection Committee

Dear Members of the Selection Committee,

It is my distinct honor to nominate Brendan Russ for the Distinguished Teaching Assistant Award. I have had the opportunity to observe and assess the teaching contributions of our TAs across multiple courses and have never felt compelled to nominate someone for this award. However, Brendan has gone above and beyond the normal TA duties and stands out as one of the most effective and impactful TAs in our department, with a sustained record of excellence in instruction, curriculum development, and student support that clearly sets him apart him from his peers.

Brendan has served as an instructor in six different chemical engineering classes over 11 quarters, most notably CH ENGR 108A/B, the senior capstone design course, which is the most demanding course across our curriculum. In this course, he has played a central role in improving the quality of instruction and the overall student experience by providing students with clear, structured guidance on how to communicate complex engineering designs at a professional level, ultimately fulfilling a long-standing curricular need. These improvements were developed thoughtfully, informed by student feedback and teaching evaluations, and have resulted in measurable improvements in student performance and confidence.

Brendan consistently implements modern, evidence-based teaching practices that emphasize active learning and student engagement. Teaching evaluations across multiple quarters show exceptionally strong scores in organization, communication, interaction, and overall effectiveness, placing him among the top TAs in our department. Student feedback repeatedly highlights his clarity, responsiveness, and ability to explain complex material in a manner that is both rigorous and accessible.

Equally important is Brendan's dedication to inclusive, student-centered teaching. He is highly attentive to differences in student preparation and learning styles and actively works to ensure that all students are empowered to succeed. Through intentional scaffolding, anonymous discussion platforms, and consistent availability outside of class, he creates learning environments in which students feel comfortable engaging with challenging material and seeking help when needed. This approach has been particularly impactful in CH ENGR 108A/B, where students must manage complex, open-ended design projects over an extended period of time.

Outside of his instructional responsibilities, Brendan has made significant contributions to curriculum development within the department and beyond. In collaboration with faculty, he co-developed a new chemical engineering technical writing course that integrates critical thinking, ethics, and discipline-specific communication. He has further strengthened his teaching expertise through formal pedagogical training focused on teaching writing to engineers and is currently contributing to a scholarly manuscript for Chemical Engineering Education that documents this curricular innovation. Through volunteering with the California NanoSystems Institute's Nanoscience Teacher Training Program, Brendan has developed new interdisciplinary experiments and improved instructional materials for K-12 teachers in the greater LA area, demonstrating his commitment to education across multiple levels.

Brendan Russ's teaching excellence has been recognized through multiple awards, including the UCLA Samueli School of Engineering Student Welfare Award and repeated departmental and student-led teaching honors. His record reflects sustained excellence, meaningful impact on student learning, and leadership in instructional improvement. I offer my strongest recommendation for his nomination for the Distinguished Teaching Assistant Award.

Sincerely,



Panagiotis D. Christofides
Distinguished Professor and Department Chair of Chemical and Biomolecular Engineering
Distinguished Professor of Electrical and Computer Engineering
William D. Van Vorst Chair in Chemical Engineering Education

DTA Self-Statement

As the oldest of five, I've been a teacher my whole life. Seeing your younger siblings glide over challenges you agonized over and climbing towards their personal goals is the most fulfilling experience there is. I strive to be an older brother for my students as well. Luckily, as a double Bruin, I am uniquely situated to connect with and inspire students using inclusive and accessible teaching strategies, fill skill holes in the curriculum by developing a technical writing module and course, and streamline their learning experience using active learning techniques.

I prioritize connecting with students in ways that encourage genuine interest and participation in the course. My SET surveys reflect the effectiveness of my personal interactions in making challenging and complex topics digestible (Teaching Artifact 1). I support the success of *all* students by filling knowledge gaps of each student individually and applying inclusive scaffolding. Because students have different levels of mastery on prerequisite concepts, I build to new concepts from many different starting points. Usually, this involves reviewing concepts from previous classes and working with students who need extra help during office hours. I also use anonymous discussion boards like Piazza, allowing students to ask and answer each other's questions without fear of judgement. I always leap at opportunities to inspire students by encouraging them to spend time on topics they relate to. When two senior design students spent hours asking me about the long-term sustainability of ethylene, I worked with them to reorganize their team so they could explore the long-term climate impact of their chemical plant. They took the lead in that part of their project and delivered a complete, industry-level life cycle analysis with recommendations to improve the sustainability of their team's project.

I improve the student learning experience by observing curriculum gaps and developing content to address them. During my undergrad, I recognized that while I had received formal training in technical writing, many of my peers had not. They struggled to write clearly and effectively under time pressure, often causing instructors to misjudge the level of understanding of the class. Now, as a TA for the very senior design course I took, I leveraged my understanding of the student experience to implement a technical writing workshop during discussion sections (Teaching Artifact 2). In response to student feedback, I also suggested technical writing content to my advisor, Prof. Carissa Eisler, who has since implemented this training into our junior lab course. I've expanded my writing pedagogy by attending Prof. Randy Fallows ENGR 495I course, which trains engineers to teach Writing-II. In collaboration with him and Prof. Jane Chang, in our department, I developed a brand-new, chemical engineering-specific Writing-II course (ENG 181W: Critical Thinking and Writing for Chemical Engineers) which will be offered in Fall 2026 (Teaching Artifact 3). We are currently writing a paper detailing how to integrate critical thinking, ethics, and technical writing into chemical engineering curriculums, and plan to publish in *Chemical Engineering Education*.

I implement modern, effective teaching strategies that emphasize critical thinking and active learning to improve learning experiences. I was first exposed to these best practices when teaching with Prof. Eisler. The incorporation of learning assistants and group work during discussion sessions enabled me to give individual attention to all ~80 students as they worked through real-world problems. I included these strategies in the design of ENGR 181W to scaffold students through recently-learned content and apply them to ethical and societal issues (Teaching Artifact 4). During this course, we have 2-hour, 20-person discussions that teach technical writing skills. Every week, we have an interactive in-class activity in which students participate in group work, think-pair-shares, and metacognitive writing exercises (Teaching Artifact 5). In

this exercise, students receive flashcard-style cutouts from a paper they read and are tasked with ordering them in a manner that makes the most sense. Then, they justify their organizational scheme to each other in small groups and bring their group decision to the class. We believe these teaching techniques will vastly improve student's communication skills in engineering classes and beyond.

As the older brother to four siblings, I spent many years working to help them avoid my mistakes without hindering their own learning and development. I will continue to develop skills in effective teaching in my work here at UCLA, and the Distinguished Teaching Award would recognize my growth as an instructor, my improvements to the chemical engineering curriculum, and my success in inspiring and encouraging the next generation of Bruins.

Bring out your inner engineer

There is an engineer in all of us. My job as an instructor is to nurture and grow the inner engineer in my students. To most, engineering is a daunting field that is too challenging and complex to understand. The challenge in teaching engineering is convincing students that they are *already* capable engineers and with hard work and perseverance, they can develop the essential skills of an engineer: problem solving, effective communication, and continuous, iterative optimization to support innovation and discovery.

Inspire Problem Solving:

Just like when a baker repeatedly modifies their cookie recipe for ideal chewiness or perfect sweetness by toying with baking times or amounts of sugar, engineers identify a process or problem, break it down into manageable parts, and iteratively improve it to optimize for specified parameters. The key to teaching engineering effectively is to connect with students' inner engineers and inspire them to improve problem-solving skills.

Encouraging curiosity is a powerful instructional skill. I encourage students to ask "why?" and empower them to satisfy their drive to discover by mirroring the reasoning processes of historical scientists and engineers, connecting their innate human curiosity to the development of key course concepts. This emboldens students to seek out answers to problems on their own, knowing the "expert," "correct" answers in the textbook are accessible to them through their problem-solving skills. I emphasize the concepts they grapple with took scientists and engineers decades to discover. It is vital to remind students that struggling with their homework problems is not a sign of inadequacy, *it is the key to their growth and development*.

To further underscore that everyone is an engineer, I help students break down problems in ways that play to their personal strengths. Every student brings their own perspective, remembers different parts of previous classes, and solves problems in unique ways. Rather than forcing students into one problem-solving method, I encourage them to take unconventional paths that align with their individual capabilities. Some students prefer analogies to the real world and connect ideas to everyday phenomena. Some students need to figure it out on their own, carefully analyzing each step of a problem like a logic puzzle. Many more just need reassurance that they can solve the problem. It is my job as an instructor to fill all those supportive roles simultaneously, providing inclusive scaffolding that empowers all students simultaneously to grow as engineers.

Impart Translational Communication Skills:

The recent collapse of public trust in science is, in part, the result of how inaccessible academic work is to general audiences. It is crucial to teach students to communicate not only within their respective fields, but also outside of academia. I impart translational communication skills that allow students to write for a variety of audiences and genres. Students feel they receive mixed messages throughout their education: write five-paragraph essays, five-paragraph essays are pedantic, do not use personal pronouns, you *must* use personal pronouns for work that *you* did. I reconcile these seemingly-conflicting instructions by reminding them that the fundamental principles of writing are all the same: identify your audience and purpose, define an appropriate structure, select an effective style, and conform to genre-specific norms. These "conflicting" messages arise simply because their instructors applied some of these principles in advance.

While students utilize these principles at every step of their writing career, they rarely translate them across genres. By learning to translate their rhetorical skills, students can adapt to new situations that require different communication strategies without feeling like they need to start from scratch. I implemented a "writing as an engineer" discussion section into the CH ENGR 108AB senior design course, dramatically improving student final reports. Because one discussion section is insufficient to teach technical writing, I have created a brand-new writing-II-accredited course (ENGR 181W) designed specifically for chemical engineering juniors that teaches critical thinking and communication. We will offer the course in Fall 2026 and highlight both traditional writing-in-the-major documents (lab reports, memos) and non-traditional communication forms for engineers (dialogues, debates, elevator pitches) to give students a wide breadth of experience.

Instill a Growth Mindset about Learning:

A growth mindset is a mentality in which the learner sees mistakes as learning opportunities, rather than failures, and believes they can improve with practice. I cultivate this environment in my courses by encouraging students to think about their own learning process (metacognition) and emphasizing that they can improve their learning processes with practice. With inflated GPAs, increasingly competitive post-graduate applications, and the temptation to use large language models to circumvent the development of critical thinking skills, students feel more pressure than ever to be perfect, while simultaneously struggling to find the motivation to learn and grow themselves. Inspiring students to become capable, confident learners is key to promoting their long-term success. In our senior design class, I ask students to rank the educational value of parts of the class and provide suggestions on how to improve the course. I help them identify which suggestions they can apply themselves, like making a cheatsheet, and which suggestions I can implement, like reading quizzes to encourage attendance. With this strategy, I demonstrate to students that they have agency over their learning process and small actions can dramatically improve their learning experience. I also require "process memos" in their senior design progress reports, where students describe the current problems and how they plan to solve them. At the end of the quarter, they compile all their process memos into a "problems and solutions" section that acts as a summary of their learning process. By requiring students to describe where they struggle, I hope to decouple the link between their struggles and their grades; I expect them to struggle with some concepts, and describing their learning process is part of their grade.

Teaching is one of my greatest joys in life. I am fulfilled by making deep, personal connections with my students, watching them achieve profound personal growth, and leaving a lasting impact on their educational journeys.

Teaching Artifact 1: SET Survey Excerpts

Collated SET Survey Scores (Scores out of 9)

Quarter	22W	22S	22F	23W	23S	23F	24W	24S	25W	25S
Course (CH ENGR)	102A	102B	101A	108A	108B	101A	108A	108B	108A	108B
Response Rate	8/14	9/30	27/30	28/45	28/45	44/50	29/45	17/45	70/76	29/45
Knowledge	8.88	8.78	8.89	8.04	8.43	8.7	8.86	8.88	8.73	8.52
Concern	8.88	8.67	8.63	8.04	8.32	8.5	8.83	8.94	8.67	8.45
Organization	8.63	8.67	8.65	8.14	8.39	8.57	8.79	8.94	8.7	8.38
Scope	8.63	8.78	8.54	8.14	8.57	8.36	8.79	8.88	8.66	8.45
Interaction	9	8.78	8.67	8.15	8.5	8.5	8.76	8.94	8.69	8.37
Communication Skills	8.88	8.78	8.56	8.18	8.54	8.59	8.76	8.94	8.66	8.31
Value	8.63	8.78	8.62	7.79	8.04	8.36	8.64	8.94	8.71	8.28
Overall	8.75	8.89	8.7	8	8.43	8.57	8.79	8.94	8.74	8.41

Capstone Kernels

“Brendan is quite literally the core pillar of this class. Without him, safe to say I would be drowning. He is so responsive and delivers information in such a clear and simple way and his office hours are the holy grail in this class. Something all of us students can probably agree on is that he is even better than the professors. Meant for teaching and it is easy to see how much the profs rely on him and the other TAs. Very inspirational, **he deserves an award**” [25W 108A]

“Brendan does too much for this class, he is an incredible TA, **supporting students both academically and with moral support in an undoubtedly challenging class**.” [25S 108B]

“Brendan **was always very concerned about student learning and understanding and would always do whatever he could to help students with project issues**. His office hours were always efficient sessions and a lot of my understanding of PRO/II modeling as well as the conceptual project understanding came from Brendan.” [24S 108B]

“Brendan knew exactly what was going on, and was very helpful. His feedback on reports was very helpful and appreciated… I would like to thank him for the incredible amount of time he dedicated to this class, and I think **he made a huge effort to make this class much more bearable for the students**.” [23S 108B]

“Brendan is not only incredibly smart and knowledgable, he is dependable and **has an amazing grasp on how students can best learn.** I feel lucky to have been able to depend on Brendan. Whatever course Brendan has been an addition to, he has always been a highlight of excellence.” [S23 108B]

Delectable Digestives

“I thought that Brendan was an amazing TA and he really helped to teach us what we were discussing in lecture at a **much more manageable and understandable level**.” [22W 102A]

“Brendan is one of the best teachers I have had at UCLA. He is extremely responsive, knows the content well, is **good at explaining it in a digestible way**, and seems to care about the students. He is helpful for both class logistics and material.” [25W 108A]

“Brendan is a great TA. He understands the class material and is able to **re-explain and contextualize it in a way that gives students a better understanding**. The discussion sections are always super organized but still feel natural and allow students an opportunity to ask about concepts they are unsure of. Brendan also will respond to email questions super fast and shows that he is concerned about student learning.” [22S 102B]

“**Very deep and thorough explanation of difficult and complex concepts**. Excellent TA who goes above and beyond.” [F22 101A]

“Brendan **does an excellent job explaining concepts and technical specifications for students**. He is very knowledgeable about the content in the class and is always open to helping students.” [F24 108A]

“Very knowledgeable, helpful, **put effort into course to create materials to expand on less understood points**.” [F23 101A]

“He … makes sure to **ask questions of his own to make sure students are engaged and understanding the material well**. This discussion itself is very helpful as it allows students to apply what they learned that week to worksheets in an environment where they can ask questions when troubled.” [F22 101A]

Approachable Accessibility

“Very knowledgeable, **very approachable, and had great communication skills. Always open to help**, very understanding, and very very kind” [24S 108B]

“Brendan is easily the best TA I have ever had in a chemical engineering course and stands out in every way, from mastery of content to communication skills. **He goes above and beyond and is so involved and available**.” [25S 108B]

“Brenden is easily the best TA I have ever had. He is extremely organized and has such an impressive mastery of the content. **I always felt comfortable asking questions, and whenever I did so, he would go above and beyond in giving an answer and additional resources**.” [23F 101A]

Teaching Artifact 2: Excerpt from Technical Writing Module for CH ENGR 108AB

Academic Writing vs Business Writing

ACADEMIC WRITING

Target Audience: Experts

Objective:

1. To share new scientific insights
2. To prove new methods
3. To suggest new ideas

BUSINESS WRITING

Target Audience: Experts, Accountants, Technicians, Executives, Press, Public

Objective: Depends!

1. To Update (progress reports, memos)
2. To Recommend (reports, memos)
3. To Instruct (SOPs, manuals)
4. To Propose (proposals, reports)
5. To Justify (reports, memos)

Academic Writing vs Business Writing – Types of Writing

ACADEMIC WRITING

Types of Writing: Papers, Studies, Proposals*

BUSINESS WRITING

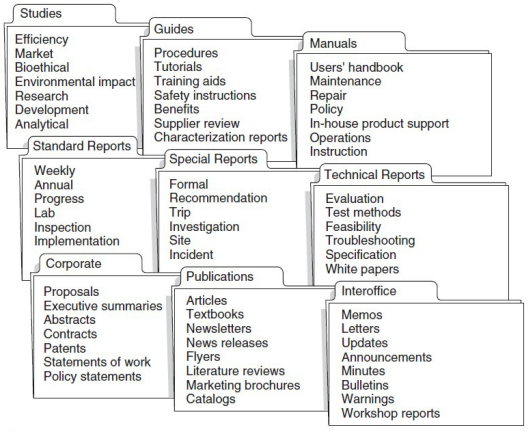


Figure 1-2 Throughout their careers, engineers write many kinds of documents in various contexts and with different purposes and audiences.

Beer, D.F. and McMurrey, D.A. (2014) *A guide to writing as an Engineer.* Hoboken, NJ: Wiley. (Page 4)

Business Writing – How to Write a Document

1. Determine the Audience
2. Determine the Objective
3. Create an Outline
4. Fill the Outline
5. Garnish (Executive Summary, Cover Letters, etc…)

Sometimes there are multiple audiences for one document, and sometimes there are different audiences for different sections of a document

Wordiness Examples

Bad: I regret to say that at this point in time I basically do not have access to that specific information…

I don't know (yet)

Small Group Exercise: Make the longest sentence possible with the same meaning.

Bad: It is our considered recommendation that a new computer should be purchased…

We recommend buying a new computer

Small Group Exercise: Make the longest sentence possible with the same meaning.

• Beer, D.F. and McMurrey, D.A. (2014) *A guide to writing as an Engineer.* Hoboken, NJ: Wiley. (Page 53)

ENGR 181W
Critical Thinking and Writing for Chemical Engineers
Course Syllabus

Instructor: Jane P. Chang (jpchang@ucla.edu)

Class Time: TR 10-11:50 am

Discussion: W 10-11:50am, F 12-1:50pm, F 1-2:50pm

Office Hours: Two 1-hour sessions for each TA (TBD). 1 hour session for instructor (TBD)

Textbook: *Own Your Perspective by Fallows and Christensen*

Attendance: In-person attendance is required.

1) Some in-class writing/exercises/quizzes are graded.
2) There will be a final grade deduction of 1% for each unexcused absence (late arrival by >15 mins counts as an absence since quiz is usually done at the start of the class).

Reading list

- Week 1: BP oil spill congressional report. Assignment: incident report and 20 quotes
- Week 2: Cobalt red. Assignment: 2 chapters.
- Week 3: The immortal life of Henrietta Lacks. Assignment: 3 chapters.
- Week 4: Emission of CFC, PFC, HFC. Assignment: One Nobel lecture and one journal article (students are assigned to specific readings based on UID).

Course Description and Learning Outcomes:

This course focuses on critical thinking and writing for chemical engineers. It aims to provide students with a broader societal context through the examination of ethical issues related to chemical engineering and help them develop skills to more effectively communicate their thoughts and their work. This course will have two major components, engineering ethics and technical writing. The engineering ethics component focuses on case study while the technical components focuses on concepts from four chemical engineering core courses. The course will provide students with tools to think more critically and holistically about the ethical and societal dimensions of their work as chemical engineers. In-class writing and exercises are designed to help students with homework, team project, debate and final lab report.

Learning outcomes emphasizing the relationship between the course content and the writing

- How to analyze writing to improve writing. The students will practice the AXES (assertions, examples, explanations, and significance) on reading assignments and their own and their peers' writing.
- How to write professional technical documents. Technical work is only useful if it is effectively communicated to key stakeholders, including peers, supervisors, funding agencies, and the public.

Learning outcomes for writing in the discipline

- How to think critically about engineering ethics, using case studies that focus on the four options in the department (chemical, bio, environmental, semiconductor)

- How to work collaboratively (in debate and team project) and individually (lab reports), to prepare for CHE 104A/B/C/D Lab Courses.

Team Project

Students will be matched by topics of interest into a team in week 2 then work to propose a project topic for review/approval by instructor. The topic should be a chemical engineering ethics topic with data that the students can perform quantitative analysis. Students can choose the case study discussed in the class as a topic, but additional and independent research is needed. The project proposal will go through a drafting and revision process in consultation with the TAs. The approved project topic will be used to structure the debate. After the debate, the team will work collaboratively to present their analysis and argument.

Due dates: The team proposal (2 pages) in week 4, revision (2 pages) in week 6, debate in week 8 and team presentation in week 10.

Individual Technical Reports

Students will write two technical lab reports: These 5-8 page technical reports follow the department's writing guide in CHE104A (including introduction, methods, results, discussion, appendix). Each report will undergo a guided peer-review process and the revised version is due for grading.

a. Lab1 Report: Lab1 provides the same dataset to all students in week 3 based on CHE100 or CHE101A course content. Students need to independently develop and write the report based on data/error analysis, comparison to theoretical models and submit the draft in week 6. In week 7, each student will submit an **Individual Reflection Piece** (approximately 250 words), **written by hand**. In narrative form, students will address the following questions: 1) What section did you need to make most changes based on the feedback? 2) What did you learn about your strengths and weaknesses as a technical writer from working with your writing and the peer review process? The revised and typeset Lab1 report is due at the same time.

b) Lab2 Report: Lab 2 provides *individualized dataset* to each student in week 7 based on CHE100 or CHE101A course content. The student needs to independently develop and write the report based on specific data set provided. The draft is due in week 9 and will be peer-reviewed and revised in a mini-conference class. In the appendix, students need to include a paragraph articulating how they implemented the feedback from Lab1 grading in the writing of Lab2 report. The individual final report is due on the University specified final exam date/time.

Homework and Analytical Reflection

Homework is assigned on Tuesday and due on Thursday of the following week. You will revise one in-class writing for the homework and can use all in-class writing/exercise/discussion in your homework.

HW#1	**Illustration of Data (figures)** Design figures/tables from BP incident report with the goal to educate/explain; revise in-class writing 4 (2 typeset pages)
HW#2	**Two Page Technical Memo** From conflict mining to a catalysis experiment; revise in-class writing 5
HW#3	**Dialog (AXES, prep for debate, prep for longer report)** From Bio ethics (right-and-right dilemma) to cell culture and fermenter; revise in-class writing 7 (2 typeset pages)
HW#4	**Paragraph (prep for full length report)** Emission of HFC on the environment; revise in-class writing 8 (3 typeset pages)

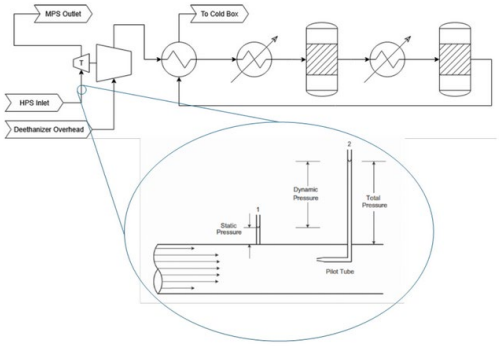
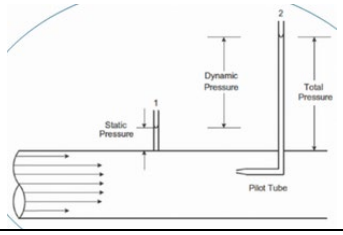
ENGR 181W Weekly Schedule

W	Lecture (4 hours)		Discussion (2 hours)
	Ethics	**Technical Writing**	**TA instruction**
1	Intro to critical thinking and writing and chemical safety	Ethical theory, nature of analysis, technical writing	Audience and purpose (library presentation on research)
2	Case Study: Chemical Safety/BP Teaming	Developing assertions from a close reading (BP case)	The craft of presenting data
3	Case Study: Conflict Mining	Writing memo based on Catalytic reactor, DOF **HW#1 due**	Structure of technical reports (focus on memo) **Lab1 data**
4	Case Study: HeLa Cells Team proposal due	Writing about fermentation and emission to C footprint **HW#2 due**	Technical writing style
5	Case Study: CFC/PFC/HFC (Nobel lecture/pub)	**E**xplanation, **S**ignificance and resources **HW#3 due**	Paragraph structure
6	Mid-term exam focused on Technical Writing	**Own** your perspective based on W 4,5,7,8 **HW#4 due**	Mini conference 1 **Lab1 draft DUE**
7	Deep Dive – team project	(Veteran's Day) **Reflection DUE; Lab1 DUE**	Effective presentation (slides comments) **Lab 2 data**
8	Debate preparation	Debate (group 1)	Debate (group 2)
9	Responsible approaches to emerging technologies	Mini conference 2 **Lab2 draft DUE**	(Thanksgiving)
10	Team presentation (group 1)	Team presentation (group 2)	Workshop for final report
F	In-person final exam on engineering ethics Lab2 Report DUE		

Grading	%
Classroom writing/exercise/participation and unannounced individual quizzes	25%
Homework	12%
Team Presentation (2% proposal, 8% presentation)	10%
Debate	10%
In-class writing assessment (aka mid-term exam)	10%
Individual technical reports (6%, 12%)	18%
In-class writing assessment (aka final exam)	15%

Penalty (zero-point assignment) will be imposed on plagiarism and AI generated content.

Teaching Artifact 4: *Chemical Engineering Education* Possible Scaffolding Figure: From Fundamentals to Ethical and Societal Issues

Scaffolding	Addition to Assignment
Connect to ethical issues, societal impact, etc… [energy use/loss, etc…]	[After scenario paragraph]: As your company's expert on fluid dynamics, the chief engineer has tasked you with understanding the data. You need to consider how to best understand the data using theory, why the recorded data (might) deviate from theory, and provide a recommendation about what to change to improve plant performance in some metric. Write a 5-8 page lab report in which you process the data, compare it to theory, and provide meaningful recommendations to improve the performance or safety of the chemical plant based on the data your technician gave you.
Connect to a realistic chemical engineering scenario [prompt]	Technicians at your ethylene chemical plant recorded the following pressure values in a high-pressure steam delivery pipe (located at the green callout on the below PFD) using a manometer. This pipe feeds a compressor that raises the pressure of process gas entering the acetylene reactors, which convert acetylene to ethylene. Because acetylene left in the process stream degrades upstream catalysts, these reactors are vital to the overall plant performances.
Connect to Learning Outcomes	Write a 5-8 page lab report in which you process the data, compare it to theory, and provide meaningful recommendations.
Where does this concept appear in later courses? [Senior Design Ethylene Plant]	The following is a process flow diagram detailing the steam system in your company's ethylene plant and a schematic showing the manometer/pipe setup technicians used to collect the pressure data: 
What do students already know? [Fluid Dynamics Week 1/2]	

Teaching Artifact 5: Organization Exercise

Instructions: Sort the following into an order that makes sense. Justify your decision.

A. **Effect of MEA concentration:** Fig. ▮ shows the overall mass transfer coefficient increases from 0.2943 to 0.4044 kmol·m^{-3}·h^{-1}·kPa^{-1} as the MEA concentration increases from 10 wt% to 40 wt% in a fixed 12 vol% CO_2 inlet concentration. This tendency holds true for other CO_2 inlet concentrations. This is attributed to the fact that the increase of the MEA concentration yields more active MEA molecules available to diffuse toward the gas-liquid surface and then react with CO_2 molecules, which enlarge the reaction enhancement factor and lead to a better mass transfer performance. Nevertheless, from the point of industrial application, the viscosity of solution increases significantly and severe corrosion would occur in the equipment at higher MEA concentration. These side effects would block the improvement of CO_2 absorption performance and increase the maintenance cost. Hence, the overall mass transfer coefficient and cost should be balanced when increasing the concentration of MEA.

B. **The gas phase mass transfer coefficient and effective contacting area:** The parameters discussed above can be summarized as the effect of liquid to gas ratio and mole ratio of MEA to CO_2. As can be seen from Fig. ▮, the overall mass transfer coefficient increases obviously with the increase of liquid to gas ratio and mole ratio of MEA to CO_2. The overall mass transfer coefficient increases from 0.1668 to 0.4519 kmol·m^{-3}·h^{-1}·kPa^{-1} as the liquid to gas ratio increases from 0.0136 to 0.0335. This is because with the increase of liquid to gas ratio, the liquid flow rate increases and the boundary layer of liquid phase become thinner, which decreased the mass transfer resistance and accelerate the reaction process. Furthermore, the increasing of liquid to gas ratio leads to an increasing of liquid to gas contacting area and this will also enhance the absorption performance. However, under the larger values of liquid to gas ratio, the growth of the overall mass transfer coefficient changes slower. Fig. ▮ provides that in a fixed liquid to gas ratio of 0.0267, the overall mass transfer coefficient increases from 0.2784 to 0.3709 kmol·m^{-3}·h^{-1}·kPa^{-1} as the mole ratio of MEA to CO_2 increases from 6.39 mole/mole to 25.5 mole/mole. The reason is that at the same liquid to gas ratio, the increase of MEA to CO_2 mole ratio allows more active MEA molecules to contact and react with CO_2 molecules, leading to the increase of overall mass transfer coefficient. It can be concluded that liquid to gas ratio and mole ratio of MEA to CO_2 are two key factors which affect the performance of CO_2 absorption process.

C. **Effect of CO_2 concentration:** The experimental results in Fig. ▮ show that the overall mass transfer coefficient decreases from 0.3326 to 0.2146 kmol·m^{-3}·h^{-1}·kPa^{-1} as the CO_2 concentration increases from 8 vol% to 18 vol% in a fixed 80 L·h^{-1} liquid flow rate. This tendency also holds true for other liquid flow rate. In general, according to the two-film theory, the gas phase driving force and gas phase mass transfer coefficient increase with the increase of CO_2 concentration, which will enhance the absorption process. Whereas, the mole ratio of MEA to CO_2 decreases with the increasing CO_2 inlet concentration, which means more CO_2 molecules react with limited active MEA molecules and this will cause the reduction of absorption performance. Moreover, the gas phase driving force $P\left(y_{CO_2}, G - y^*_{CO_2}\right)$ increased with the increase of CO_2 concentration, which will also lead to the decrease of the overall mass transfer coefficient.

D. **Effect of gas flow rate:** Fig. ▮ shows that the overall mass transfer coefficient increases from 0.1432 to 0.4773 kmol·m^{-3}·h^{-1}·kPa^{-1} as the gas flow rate increases from 1.0 m^3·h^{-1} to 5.0 m^3·h^{-1} in a fixed 12 vol% CO_2 inlet concentration. This tendency also holds true for other CO_2 inlet concentrations. According to the gas-liquid mass transfer theory, the mass transfer coefficient increases with the increase of gas flow rate. This is because as the total gas flow rate increased, the amount of CO_2 molecules available to diffuse toward the gas-liquid surface and then contact and react with MEA molecules increased, which will lead to an increase of the overall mass transfer coefficient.

E. **Effect of liquid flow rate:** As can be seen from Fig. ▮, the overall mass transfer coefficient increases with liquid flow rate in varies CO_2 inlet concentrations. In a fixed 12 vol% CO_2 inlet concentration, the overall mass transfer coefficient increases from 0.2837 to 0.3833 kmol·m^{-3}·h^{-1}·kPa^{-1} as the liquid flow rate increases from 60 L·h^{-1} to 120 L·h^{-1}. This is because with the increase of liquid flow rate, the droplets flow rate increases and the boundary layer of liquid phase decreases. So the resistance for gas diffusion to the liquid phase decreased and the mass transfer performance is enhanced. As a result, the overall mass transfer coefficient increases with the liquid flow rate. However, the increasing tendency dropped rapidly at the higher range of liquid flow rate, this is because the reduction in droplet size by the increasing liquid flow rate becomes insignificant and the increase of effective interfacial area is limited. Hence, the mass transfer performance can not be enhanced furthermore at higher liquid flow rate.

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name

Briefly describe how you know the nominee. (50 words max)

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name: Alexander Geldmeyer

Briefly describe how you know the nominee. (50 words max)

Brendan was my TA in 2024, when I was a student in the chemical engineering senior design capstone class (CH ENGR 108A and 108B). Over the past two years, I have had the pleasure of working alongside him as a fellow TA.

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Brendan creates a great learning environment where students feel comfortable asking questions and feel welcome to reach out via email or come to office hours. He brings genuine enthusiasm to discussion sections, making them not only informative but also fun. He clearly cares about the students' learning and connecting with them on a relaxed and friendly level.

Furthermore, getting help from Brendan is encouraging and motivating. He never just gives out the answer. He walks students through the problem-solving process. He breaks down the process and asks questions, so students can get to the answer by themselves. While getting help from Brendan, his patience and encouragement make it feel more like you are working with him rather than getting help from him. You brainstorm together, but the students pave their own way to the solution. It's encouraging that he doesn't just approve of your answer being correct, but also of your problem-solving process. As a student, this is reassuring and gives you confidence because next time the problem comes around, you actually know how to solve it by yourself.

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Brendan's discussion sections are always very well organized and relevant. He adapts the discussion sections to what students are struggling with the most. If a particular concept in class was not explained as clearly, or if many students are asking questions about it, he dedicates a discussion section to explaining that concept. He does this by including live demonstrations and making sure to take pauses for questions. His ability to answer questions goes beyond just being good at explaining problems. When he notices that a student didn't fully understand his answer, he goes over it again but breaks the concepts down into smaller pieces, making sure the student can follow each step. Brendan is extremely knowledgeable about the concepts that he teaches. He takes the time to understand and learn the concepts before he teaches them. Allowing him to answer any question students ask of him.

On top of his great discussion sections, Brendan always makes himself available for questions. He quickly responds to emails and ensures that his office hours don't overlap with other required classes. Brendan puts the students first and ensures that they can learn to the best of their abilities.

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

Alongside Brendan's amazing teaching abilities, he also focuses on developing new TAs. Brendan puts in extra effort to catch up new TAs on how the class is run and organized. He also provides advice and resources to the new TAs to be best prepared for teaching and grading class material. Last year, when I TAed alongside him, he provided me with a list of good references to review to get a good understanding of the material I would be teaching. I was better prepared and felt more comfortable answering student questions.

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name

Briefly describe how you know the nominee. (50 words max)

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

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To: Distinguished Teaching Award Selection Committee February 7, 2026

Brendan Russ is a shiny example of how an effective teaching assistant can enhance the course and curriculum development and enrich the learning experience of many students. As UCLA celebrates distinguished teaching assistants, I most enthusiastically recommend Brendan Russ this recognition.

When I decided to develop the "Critical Thinking and Writing for Chemical Engineers" course in Fall 2025, I know exactly whom I want to work with – Brendan Russ. I know this is not an easy ask, as Brendan would have to work on course development with me and take a Writing II TA training course (with Randy Fallows, 2013 UCLA Distinguished Teaching Awardee), on top of his own productive research. I explained this fully and was delighted when Brendan responded positively. He gave me a genuine reason – he knows our curriculum well and agrees this course will help all our undergraduate students, including his youngest sibling who just started as a freshman in our department.

There is a writer in all of us. To bring out the writer in an engineer, Brendan and I mapped out our core courses where foundation knowledge can be leveraged to support critical thinking and writing. We co-developed this course to provide students with a broader societal context through the examination of ethical issues related to chemical engineering and help them develop skills to more effectively communicate their thoughts and technical work. This course has two major components, engineering ethics and technical writing. The former builds upon case studies related to the four options in our curriculum, and the latter focuses on concepts from four chemical engineering core courses. Brendan and I met weekly and developed a course planner that included in-class writing and exercises, homework, scaffolding the mid-term and final writing assignments. He also contributed significantly to the course survey that we jointly developed with UCLA Teaching and Learning Center. This course, ENGR 181W, was recently approved by the Writing II committee, for its first offering in Fall 2026.

The Pattern. Brendan Russ demonstrates an exquisite pattern of effective teaching. He is not satisfied with just incorporating current best practices. He innovates and creates new practices that others can follow. An example is the scaffolding of the final individual writing assignment in "Critical Thinking and Writing for Chemical Engineers." Brendan developed an assignment guide that connects mass and energy balance in the first CHE course (CHE100) to the process flow diagrams that are central to our capstone courses (CHE108A/B). He is currently preparing a first-authored manuscript for submission to Chemical Engineering Education, a premier journal in our discipline.

Engineers can be good at numbers and words. There is often a misconception, even a joke, that engineers are only good at numbers. Brendan proves this untrue through his effectiveness in guiding students at all levels to improve. The new course he helped develop is a critical link in our curriculum that will help more engineering students to excel at numbers and words.

I most strongly recommend Brendan Russ for UCLA Distinguished Teaching Assistant Award.

Best regards,



Jane P. Chang
Distinguished Professor and William F. Seyer Chair